Homework 5

Graded

Student

Colin Skinner

Total Points

100 / 100 pts

Question 1

Problem 1 — **25** / 25 pts

1.1 a — **8** / 8 pts

✔ **– 0 pts** Correct

– 1 pt Minor Setup/Math Error

– 3 pts Major Setup/Math Error

– 8 pts No attempt

1.2 b — **5** / 5 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error / Missing Final Result

– 5 pts No attempt

1.3 c — **6** / 6 pts

✔ **– 0 pts** Correct

– 2 pts Missing/Incorrect conclusion

– 3 pts Missing/Incorrect plot

– 6 pts No attempt

1.4 d — **6** / 6 pts

✔ **– 0 pts** Correct

– 1 pt Minor Plotting Error

– 3 pts Incorrect Plot

– 5 pts No plot shown

– 6 pts No attempt

Question 2

Problem 2 **20** / 20 pts

2.1 **a** **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

– 2 pts Integration Error

– 3 pts Force Balance Error

– 5 pts Did not complete the integral

– 7 pts Did not set up the integral / Incorrect Setup

– 10 pts No attempt

2.2 **b** **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

– 2 pts Incorrect Boundary Condition

– 3 pts Missing final answer

– 4 pts Major Error

– 10 pts No attempt

Question 3

Problem 3 — **30** / 30 pts

3.1 a — **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** Minor Error
- **– 2 pts** Integration mistakes
- **– 2 pts** Does not apply / incorrect initial condition for one integration
- **– 4 pts** Does not apply / incorrect initial condition for multiple integrations
- **– 4 pts** Setup Error
- **– 8 pts** did not solve for x and y as a function of time
- **– 15 pts** No attempt

3.2 b — **15** / 15 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** Minor Error
- **– 2 pts** Integration mistakes
- **– 2 pts** Does not apply / incorrect initial condition for one integration
- **– 4 pts** Does not apply / incorrect initial condition for multiple integrations
- **– 4 pts** Setup Error
- **– 15 pts** No attempt

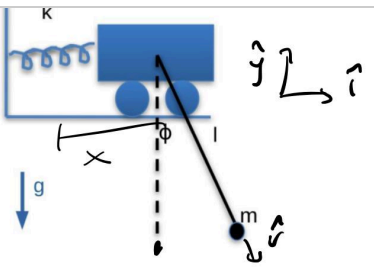
Question 4

Problem 4 — **25** / 25 pts

- ✔ **– 0 pts** Correct
- **– 2 pts** Errors or missing Lagrange Multiplier for one EOM
- **– 1 pt** Small Error
- **+ 15 pts** Correct Lagrangian and contraint equation
- **– 25 pts** No attempt

Question assigned to the following page: 1.1

t length of



a. Find EOMs by Lagrange method and discuss the physical significance any of them might have
b. Find the frequency of the oscillating pendulum for small motions about the vertical when the cart is fixed.
c. Use Matlab to find the frequency of the oscillating pendulum when the cart is fixed. DO NOT simplify the EOM by making the small angle assumption. Plot ϕ vs. time, and find the frequency of oscillation for the following initial conditions: $\phi'(0) = 0$, $\phi(0) = 5°, 10°, 15°, 30°, 60°$. Compare the frequencies of oscillation found to the result of part b (you can do this graphically). For what $\phi(0)$ is the small angle assumption accurate? Use numerical values: $m = 1kg$, $l = 2$ meters, $k = 5$ N/m, $M = 2$ kg.
d. Use MATLAB to simulate the full system when the cart is NOT fixed. Use initial cart conditions: $x(0) = 0$, $x'(0) = 0.5m/s$. Plot ϕ vs. time for the ϕ initial conditions from part c. Compare these plots to those of part (c). Use same numerical values as part (c).

Coords: ϕ, x

$$\vec{r}_p = (x + l\sin\phi)\hat{i} - l\cos\phi\,\hat{j}$$

$$\vec{v}_p = (\dot{x} + l\dot{\phi}\cos\phi)\hat{i} + l\dot{\phi}\sin\phi\,\hat{j}$$

$$T_p = \frac{1}{2}m\left((\dot{x} + l\dot{\phi}\cos\phi)^2 + l^2\dot{\phi}^2\sin^2\phi\right)$$

$$= \frac{1}{2}m\left(\dot{x}^2 + 2\dot{x}\dot{\phi}l\cos\phi + l^2\dot{\phi}^2\cos^2\phi + l^2\dot{\phi}^2\sin^2\phi\right)$$

$$= \frac{1}{2}m\left(\dot{x}^2 + 2\dot{x}\dot{\phi}l\cos\phi + l^2\dot{\phi}^2\right)$$

$$T_c = \frac{1}{2}M\dot{x}^2 \rightsquigarrow T = \frac{1}{2}m\dot{x}^2 + m\dot{x}\dot{\phi}l\cos\phi + \frac{1}{2}ml^2\dot{\phi}^2 + \frac{1}{2}M\dot{x}^2$$

$$= \frac{1}{2}(m+M)\dot{x}^2 + m\dot{x}\dot{\phi}l\cos\phi + \frac{1}{2}ml^2\dot{\phi}^2$$

$$V_c = 0 \quad V_p = -mgl\cos\phi + \frac{1}{2}kx^2$$

$$\mathcal{L} = \frac{1}{2}(m+M)\dot{x}^2 + m\dot{x}\dot{\phi}l\cos\phi + \frac{1}{2}ml^2\dot{\phi}^2 + mgl\cos\phi - \frac{1}{2}kx^2$$

$$\frac{\partial\mathcal{L}}{\partial\phi} = -m\dot{x}\dot{\phi}l\sin\phi - mgl\sin\phi$$

$$\frac{\partial\mathcal{L}}{\partial\dot{\phi}} = m\dot{x}l\cos\phi + ml^2\dot{\phi}$$

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\phi}} = m\ddot{x}l\cos\phi - m\dot{x}l\dot{\phi}\sin\phi + ml^2\ddot{\phi}$$

$$m\ddot{x}l\cos\phi - m\dot{x}l\dot{\phi}\sin\phi + ml^2\ddot{\phi} + m\dot{x}\dot{\phi}l\sin\phi + mgl\sin\phi = 0$$

$$\ddot{x}\cos\phi + l\ddot{\phi} + g\sin\phi = 0$$

$$\frac{\partial\mathcal{L}}{\partial x} = -kx \qquad \frac{\partial\mathcal{L}}{\partial\dot{x}} = (m+M)\dot{x} + m\dot{\phi}l\cos\phi$$

Questions assigned to the following page: 1.1, 1.2, and 1.3

$m\dot{\phi}^2 \ell \sin\phi$

$$(m+M)\ddot{x} + m\ddot{\phi}\ell\cos\phi - m\dot{\phi}^2\ell\sin\phi + kx = 0$$

b) $m\ddot{x}\ell\cos\phi + m\ell^2\ddot{\phi} + mg\ell\sin\phi = 0$ ($\ddot{x}=0$, $\phi \approx 0$)

simple harmonic motion

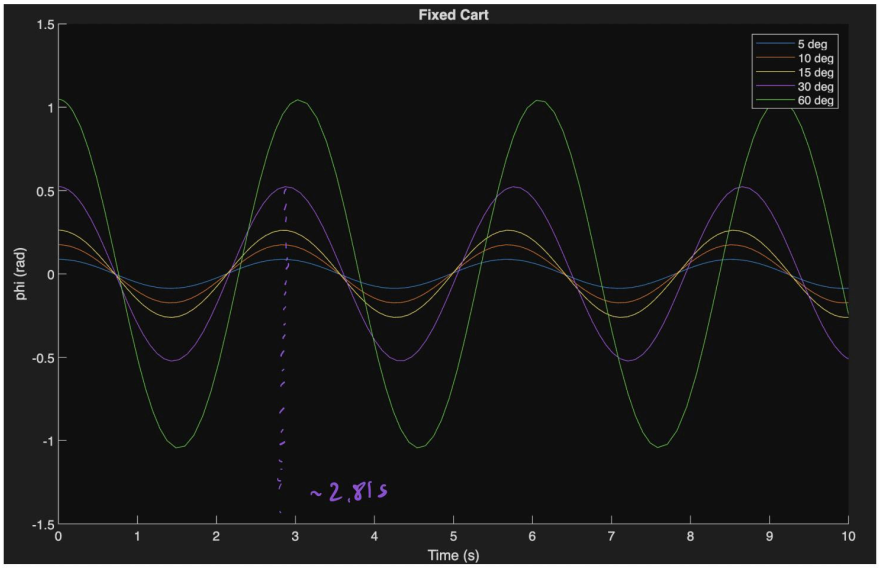
$$\ell\ddot{\phi} + g\phi = 0 \rightarrow \ddot{\phi} + \frac{g}{\ell}\phi = 0 \qquad \ddot{\phi} + \omega^2\phi = 0 \quad \therefore \quad \omega = \sqrt{\frac{g}{\ell}}$$

c) $\ddot{x}\cos\phi + \ell\ddot{\phi} + g\sin\phi = 0$ ($\ddot{x}=0$)

$$\ell\ddot{\phi} + g\sin\phi = 0 \rightarrow \ell\ddot{\phi} + g\sin\phi = 0$$

if $y_1 = \dot{\phi}$ and $y_2 = \phi \rightarrow \dot{y}_1 = \ddot{y}_2 \quad \dot{y}_1 = -\frac{g}{\ell}\sin y_2$

$$\omega = \sqrt{\frac{g}{\ell}} = \sqrt{\frac{9.81}{2m}} = 2.21\ \frac{rad}{s} \rightarrow T = 2.84\ s$$



Also accurate until $\phi(0) \approx 15°$

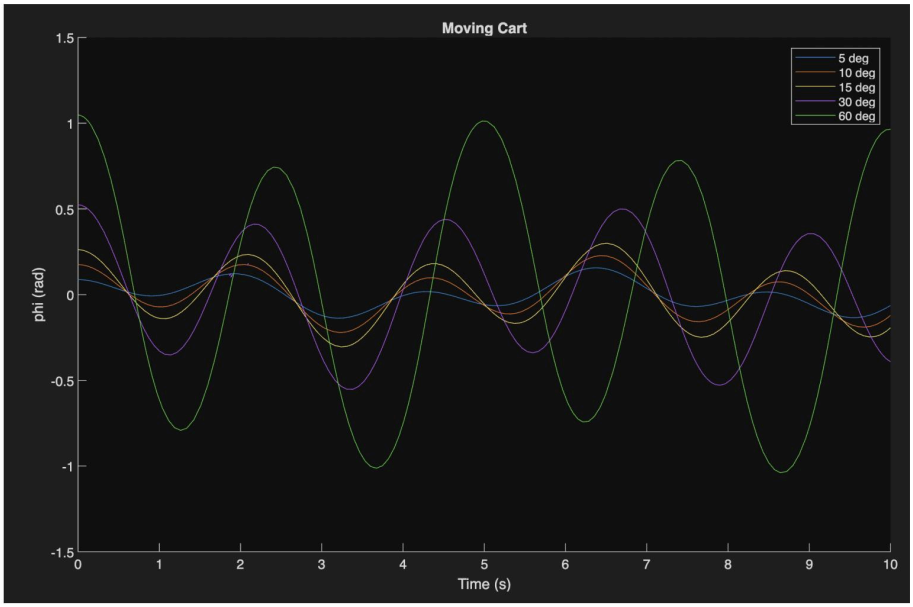
Question assigned to the following page: 1.4

$y_3 = \dot{\phi} \quad y_4 = \phi$

$\ddot{x}\cos\phi + \ell\ddot{\phi} + g\sin\phi = 0 \rightarrow$

$(m+M)\ddot{x} + m\dot{\phi}\ell\cos\phi - m\dot{\phi}^2\ell\sin\phi + kx = 0$

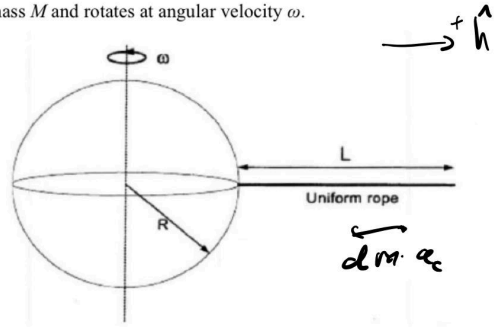
$$\begin{bmatrix} \cos\phi & \ell \\ m+M & m\ell\cos\phi \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} -g\sin\phi \\ m\dot{\phi}^2\ell\sin\phi - kx \end{bmatrix}$$



Period is not as constant for higher angles

Questions assigned to the following page: 2.1 and 2.2

…vator" …orbit around the earth above the equator. You could send an elevator up this rope to launch objects into space at less cost than required for shuttle flights. Imagine such a rope just reached the Earth's surface. Assume the rope has length L, mass m, and that the Earth has radius R and mass M and rotates at angular velocity ω.



a. Find an expression for the tension in the rope as a function of height, y, off the Earth's surface.

b. What length, L, allows the rope to hang freely (i.e. without being attached to the Earth's surface)?

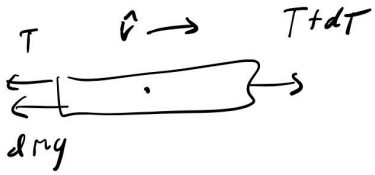
$\vec{r} = (y+R)\hat{h}$

$\vec{v} = \dot{y}\hat{h} + (y+R)(\vec{\omega}\times\hat{h})$ (0 for a point)

$\vec{v} = (y+R)\omega\hat{\theta}$

$\vec{a} = (y+R)\omega(\vec{\omega}\times\hat{\theta})$

$= -(y+R)\omega^2\hat{r}$



$\Sigma F_r = T + dT - T - dm\,g = dm\,a \rightarrow dT - dm\,g = -dm(y+R)\omega^2$

$\int_{T(y)}^{T(L)} dT = \int g - (y+R)\omega^2\, dm = \int \frac{GM}{(y+R)^2} - (y+R)\omega^2\, dm$ $\qquad \lambda = \frac{dm}{dy}$

$dm = \lambda dy$

$= \lambda GM\int_y^L \frac{1}{(y+R)^2}\,dy - \lambda\omega^2\int_y^L (y+R)\,dy$

$\frac{1}{2}y^2 + yR + \frac{1}{2}R^2$

$= \lambda GM(y+R)^{-1}(-1)\Big|_y^L - \lambda\omega^2\left[\frac{1}{2}\right](y+R)^2\Big|_y^L$

$T(L) - T(y) = -\lambda GM\left[\frac{1}{L+R} - \frac{1}{y+R}\right] - \frac{\lambda\omega^2}{2}\left[(L+R)^2 - (y+R)^2\right]$ ($T(L) = 0$ at end)

$$T(y) = \frac{mGM}{L}\left[\frac{1}{L+R} - \frac{1}{y+R}\right] + \frac{m\omega^2}{2L}\left[(L+R)^2 - (y+R)^2\right]$$

Hangs freely when $T(0) = 0$

$L^2 + 2LR + R^2 - R^2$

$T(0) = 0 = \frac{mGM}{L}\left[\frac{1}{L+R} - \frac{1}{R}\right] + \frac{m\omega^2}{2L}\left[(L+R)^2 - R^2\right]$

ignore (singularity at $L=0$)

Question assigned to the following page: 2.2

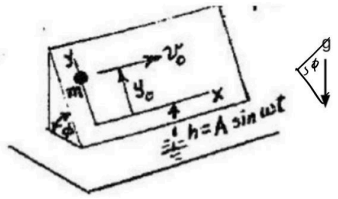
$$\frac{GM}{L+R} - \frac{GM}{R^2} + \frac{\omega^2}{2}\left(L^2 + 2RL\right) = 0$$

Wolfram Alpha

$$L = \frac{\sqrt{R^2\omega^2\left(8GM + R^3\omega^2\right) - 3R^2\omega^2}}{2R\omega^2}$$

Question assigned to the following page: 3.1

...he x axis ...ntal. In ...e a given reference level is $h = A\sin(\omega t)$.

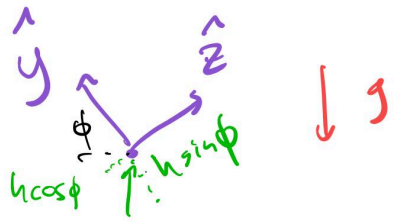


If the particle has the initial conditions, $y(0) = y_0$, $\dot{y}(0) = 0$, $x(0) = x_0$, $\dot{x}(0) = v_0$, solve for the its position (x,y) as a function of time using the following:

a. Force balance.

b. Lagrange's equation with Lagrange multipliers.

a)



$$z = 0$$

$$\ddot{h} = -A\omega^2\sin\omega t$$

$$\vec{r} = h\hat{h} + y\hat{y} = A\sin\omega t\,\hat{h} + y\hat{y}$$

$$\vec{v} = \dot{h}\hat{h} + \dot{y}\hat{y} \qquad \vec{a} = \ddot{h}\hat{h} + \ddot{y}\hat{y} = -A\omega^2\sin\omega t\left(\sin\phi\,\hat{y} + \cos\phi\,\hat{z}\right) + \ddot{y}\hat{y}$$

$$\Sigma F_y = ma_y = -mg\sin\phi = m\left(-A\omega^2\sin\omega t\sin\phi + \ddot{y}\right) =$$

$$-g\sin\phi = -A\omega^2\sin\omega t\sin\phi + \ddot{y}$$

$$\ddot{y} = \frac{d\dot{y}}{dt} = A\omega^2\sin\omega t\sin\phi - g\sin\phi$$

$$\int_0^{\dot{y}} d\dot{y} = \int_0^t A\omega^2\sin\omega t\sin\phi - g\cos\phi\,dt = A\omega^2\sin\phi\int_0^t \sin\omega t\,dt - gt\sin\phi$$

$$\dot{y} = -A\omega\sin\phi\left[\cos(\omega t) - \cos(0)\right] - gt\sin\phi$$

$$= -A\omega\sin\phi\cos\omega t + A\omega^2\sin\phi - gt\sin\phi$$

$$\int_{y_0}^{y} dy = \int_0^t -A\omega\sin\phi\cos\omega t + A\omega^2\sin\phi - gt\sin\phi\,dt$$

$$y - y_0 = -A\sin\phi\left(\sin\omega t - \sin 0\right) + A\omega^2 t\sin\phi - \frac{1}{2}gt^2\sin\phi$$

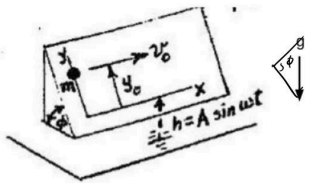
$$y = -\frac{1}{2}g\sin\phi\,t^2 + A\omega^2 t\sin\phi - A\sin\phi\sin\omega t + y_0$$

Coast cycle

$$\dot{x} = v_0 = \frac{dx}{dt} \qquad x - x_0 = v_0 t \qquad x = v_0 t + x_0$$

Question assigned to the following page: 3.2

reference level is $h = A\sin(\omega t)$.



If the particle has the initial conditions, $y(0) = y_0$, $\dot{y}(0) = 0$, $x(0) = x_0$, $\dot{x}(0) = v_0$, solve for the particle's position (x,y) as a function of time using the following:

a. Force balance.

b. Lagrange's equation with Lagrange multipliers.

b) coordinates x, y, z

constraint $f(z) = z = 0$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ $\dot{h} = A\omega\cos\omega t$

$\vec{r} = h(\sin\phi\hat{j} + \cos\phi\hat{k}) + y\hat{j} + x\hat{i} + z\hat{k}$

$\vec{r} = (y + h\sin\phi)\hat{j} + h\cos\phi\hat{k} + x\hat{i} + z\hat{k}$

$\vec{v} = (\dot{y} + \dot{h}\sin\phi)\hat{j} + (\dot{h}\cos\phi + \dot{z})\hat{k} + \dot{x}\hat{i}$ $v^2 = (\dot{y} + \dot{h}\sin\phi)^2 + (\dot{h}\cos\phi + \dot{z})^2 + \dot{x}^2$

$V = mg(h + y\sin\phi) = mg(A\sin\omega t + y\sin\phi)$ $\dot{h}^2\cos^2\phi + \dot{x}^2$

$T = \frac{1}{2}m\left[(\dot{y} + \dot{h}\sin\phi)^2 + (\dot{h}\cos\phi + \dot{z})^2 + \dot{x}^2\right]$ $\frac{\partial f}{\partial x} = 0$ $\frac{\partial f}{\partial y} = 0$ $\frac{\partial f}{\partial z} = 1$

$L = \frac{1}{2}m\left[(\dot{y} + \dot{h}\sin\phi)^2 + (\dot{h}\cos\phi + \dot{z})^2 + \dot{x}^2\right] - mg(A\sin\omega t + y\sin\phi)$

$= \frac{1}{2}m\left[(\dot{y} + A\omega\cos\omega t\sin\phi)^2 + (A\omega\cos\omega t\cos\phi + \dot{z})^2 + \dot{x}^2\right] - mg(A\sin\omega t + y\sin\phi)$

$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$ $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{x}$ $m\ddot{x} = 0$ $\dot{x} = 0$ → derived earlier $x(t) = v_0 t + x_0$

$\frac{\partial L}{\partial \dot{y}} = \frac{1}{2}m\left(2[\dot{y} + A\omega\cos(\omega t)\sin\phi]\right) = m(\dot{y} + A\omega\sin\phi\cos(\omega t))$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{y}} = m(\ddot{y} - A\omega^2\sin\phi\sin\omega t)$ $\frac{\partial L}{\partial y} = -mg\sin\phi$

$m(\ddot{y} - A\omega^2\sin\phi\sin\omega t) + mg\sin\phi = 0$ $\ddot{y} - A\omega^2\sin\phi\sin\omega t + g\sin\phi = 0$

$y(t)$ derived earlier → $y(t) = -\frac{1}{2}g\sin\phi t^2 + A\omega^2 t\sin\phi - A\sin\phi\sin\omega t + y_0$

$$y = -\frac{1}{2}g\sin\phi t^2 + A\omega^2 t\sin\phi - A\sin\phi\sin\omega t + y_0$$

Question assigned to the following page: 3.2

$\dot{z}$ z $\left(\right.$ $s\phi + \dot{z}\Big) = m\left(A\omega\cos\phi\cos\omega t + \dot{z}\right)$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = m\left(-A\omega^2\cos\phi\sin\omega t + \ddot{z}\right)$ $\quad \frac{\partial L}{\partial z} = 0$

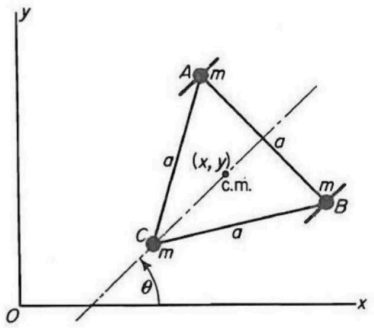
cosfunct

$m\left(-A\omega^2\cos\phi\sin\omega t + \cancel{\ddot{z}}^{0}\right) + \mu(1) = 0$

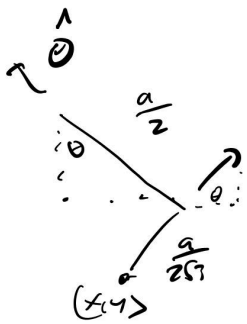
$\mu = Am\omega^2\sin\phi\sin\omega t$ Normal force to keep $z = 0$

Question assigned to the following page: 4

an equilateral
n roll without
slipping on a horizontal floor. The wheels at A and B revolve about a common axle AB and permit the motion of these two particles in a direction perpendicular to AB. A castered wheel at C allows its motion in any direction consistent with the other constraints.



Using coordinates (x,y) to specify the CM position, and coordinate θ to specify the orientation of the plane of the wheel at A and B with respect to the horizontal x-axis, obtain the equations of motion using the method of Lagrange multipliers.



$\hat{r} = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

$\hat{\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$

$\vec{v}_{cm} = \dot{x}\hat{x} + \dot{y}\hat{y} \quad \vec{v}_m = \dot{x}\hat{x} + \dot{y}\hat{y}$

$\vec{r}_{A/cm} = \frac{a}{2\sqrt{3}}\hat{r} + \frac{a}{2}\hat{\theta}$

$\vec{v}_{A/cm} = \frac{a}{2\sqrt{3}}\dot{\hat{r}} + \frac{a}{2}\dot{\hat{\theta}} = \frac{a}{2\sqrt{3}}\dot{\theta}\hat{\theta} - \frac{a}{2}\dot{\theta}\hat{r}$

for f: $\vec{v}_A = \vec{v}_{cm} + \vec{v}_{A/cm} = \dot{x}\hat{x} + \dot{y}\hat{y} + \frac{a}{2\sqrt{3}}\dot{\theta}\hat{\theta} - \frac{a}{2}\dot{\theta}\hat{r}$

$\vec{v}_A \cdot \hat{\theta} = 0 = (\dot{x})(-\sin\theta) + (\dot{y})(\cos\theta) + \left(\frac{a}{2\sqrt{3}}\dot{\theta}\right)(1) = 0$

$f = -\dot{x}\sin\theta + \dot{y}\cos\theta + \frac{a}{2\sqrt{3}}\dot{\theta} = 0$ semi-holonomic

$\vec{r}_{B/cm} = \frac{a}{2\sqrt{3}}\hat{r} - \frac{a}{2}\hat{\theta} \rightarrow \vec{v}_{B/cm} = \frac{a}{2\sqrt{3}}\dot{\theta}\hat{\theta} + \frac{a}{2}\dot{\theta}\hat{r}$

$\vec{r}_{C/cm} = -\frac{a}{\sqrt{3}}\hat{r} \quad \vec{v}_{C/cm} = -\frac{a}{\sqrt{3}}\dot{\theta}\hat{\theta}$

$T = \frac{1}{2}(3m)\left(\dot{x}^2 + \dot{y}^2\right) + 2\left(\frac{1}{2}m\left(\frac{a^2}{12}\dot{\theta}^2 + \frac{a^2}{4}\dot{\theta}^2\right)\right) + \frac{1}{2}m\frac{a^2}{3}\dot{\theta}^2$

$= \frac{3}{2}m(\dot{x}^2 + \dot{y}^2) + Ma^2\dot{\theta}^2\left(\frac{1}{12} + \frac{3}{12}\right) + \frac{1}{6}ma^2\dot{\theta}^2$

$= \frac{3}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{1}{2}ma^2\dot{\theta}^2$ NO V

$L = \frac{3}{2}m(\dot{x}^2 + \dot{y}^2) + \frac{1}{2}ma^2\dot{\theta}^2$

$\frac{\partial L}{\partial \dot{x}} = 3m\dot{x} \quad \frac{d}{dt}\left[\frac{\partial L}{\partial \dot{x}}\right] = 3m\ddot{x} \quad \frac{\partial L}{\partial x} = 0 \quad \frac{\partial f}{\partial \dot{x}} = -\sin\theta$

$3m\ddot{x} = 0$ $\boxed{3m\ddot{x} - \mu\sin\theta = 0}$

Question assigned to the following page: 4

$\frac{\partial}{\partial \dot{y}} = 3m\dot{y}$ $\quad \frac{d}{dt}\frac{\partial}{\partial \dot{y}} = 3m\ddot{y}$ $\quad \frac{\partial}{\partial y} = 0$ $\quad \frac{\partial f}{\partial \dot{y}} = -\cos\theta$

$$3m\ddot{y} - \mu\cos\theta = 0$$

$\frac{\partial L}{\partial \dot{\theta}} = ma^2\dot{\theta}$ $\quad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = ma^2\ddot{\theta}$

$\frac{\partial f}{\partial \theta} = \frac{a}{2\sqrt{2}}$

$$ma^2\ddot{\theta} + \frac{a}{2\sqrt{2}}\mu = 0$$

Question assigned to the following page: 1.3

```
close all
clear all
clc
```

```
omega: 2.21e+00 rad/s
period: 2.84e+00 s
```

Part c

```
function [x_dot] = cart_fixed_ode(t, x, l, g)
    theta_dot = x(1);
    phi = x(2);

    theta_dot_dot = - g / l * sin(phi);


    x_dot = [theta_dot_dot; theta_dot];
end
```

```
%                                phi_dot, phi
cart_fixed_initial_conditions_1 = [  0;     0  ];
cart_fixed_initial_conditions_2 = [  0;     5*pi/180  ];
cart_fixed_initial_conditions_3 = [  0;     10*pi/180  ];
cart_fixed_initial_conditions_4 = [  0;     15*pi/180  ];
cart_fixed_initial_conditions_5 = [  0;     30*pi/180  ];
cart_fixed_initial_conditions_6 = [  0;     60*pi/180  ];

tspan = [0, 10];

m = 1; % kg
l = 2; % m
k = 5; % N/m
M = 2; % kg
g = 9.81; % m/s2

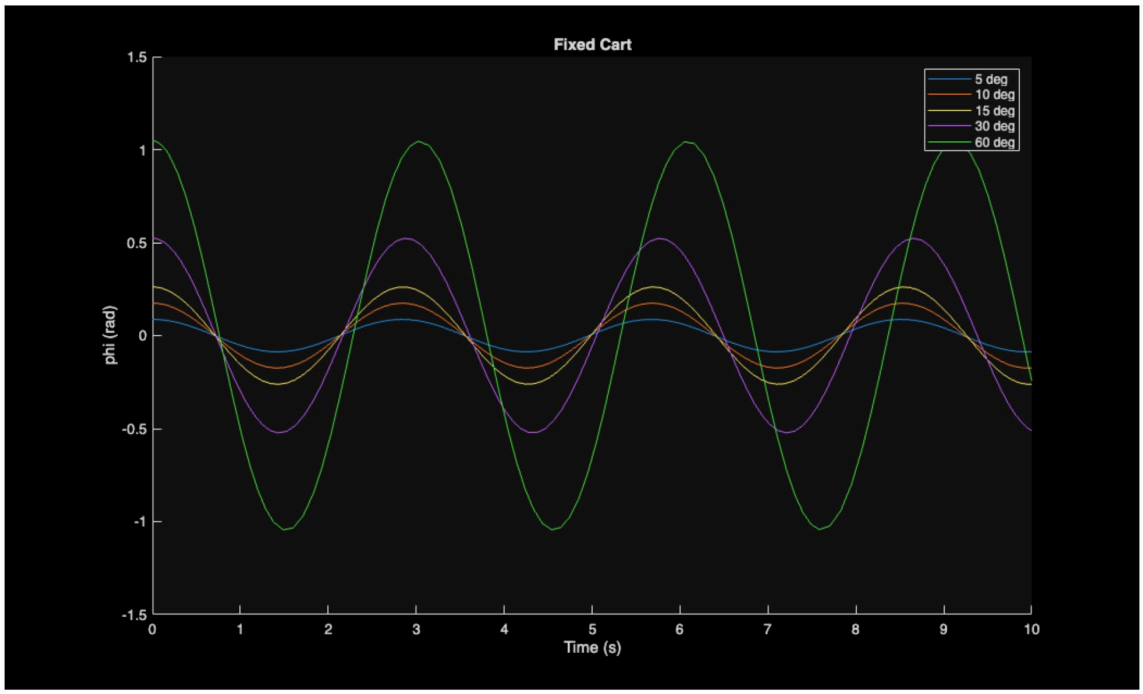
fprintf("omega: %.2d rad/s\n", sqrt(g/l))
fprintf("period: %.2d s\n", 2 * pi / sqrt(g/l))

[t1,y1] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_1);
[t2,y2] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_2);
[t3,y3] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_3);
[t4,y4] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_4);
[t5,y5] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_5);
[t6,y6] = ode45(@(t,y) cart_fixed_ode(t, y, l, g), ...
    tspan, cart_fixed_initial_conditions_6);
```

Questions assigned to the following page: 1.3 and 1.4

```
figure(1)
hold on
plot(t2, y2(:,2))
plot(t3, y3(:,2))
plot(t4, y4(:,2))
plot(t5, y5(:,2))
plot(t6, y6(:,2))
hold off
title("Fixed Cart")
legend('5 deg', '10 deg', '15 deg', '30 deg', '60 deg')
xlabel("Time (s)")
ylabel("phi (rad)")
```

```
omega: 2.21e+00 rad/s
period: 2.84e+00 s
```



Part d

```
function dydt = cart_moving_ode(t, y, l, g, m, M, k)
    x_dot = y(1); x = y(2);
    phi_dot = y(3); phi = y(4);

    A = [cos(phi), l;   m + M, m*l*cos(phi)];

    b = [-g * sin(phi); m * phi_dot^2*l*sin(phi) - k*x];

    dot_dot_vec = A\b;  % solve for [x_ddot; phi_ddot]
    x_dot_dot = dot_dot_vec(1);
```

Question assigned to the following page: 1.4

```
    phi_dot_dot = dot_dot_vec(2);

    dydt = [x_dot_dot; x_dot; phi_dot_dot; phi_dot];
end

%                                    x_dot, x,phi_dot, phi
cart_moving_initial_conditions_1 = [  0.5; 0; 0;     0  ];
cart_moving_initial_conditions_2 = [  0.5; 0; 0;     5*pi/180  ];
cart_moving_initial_conditions_3 = [  0.5; 0; 0;     10*pi/180  ];
cart_moving_initial_conditions_4 = [  0.5; 0; 0;     15*pi/180  ];
cart_moving_initial_conditions_5 = [  0.5; 0; 0;     30*pi/180  ];
cart_moving_initial_conditions_6 = [  0.5; 0; 0;     60*pi/180  ];

tspan = [0, 10];

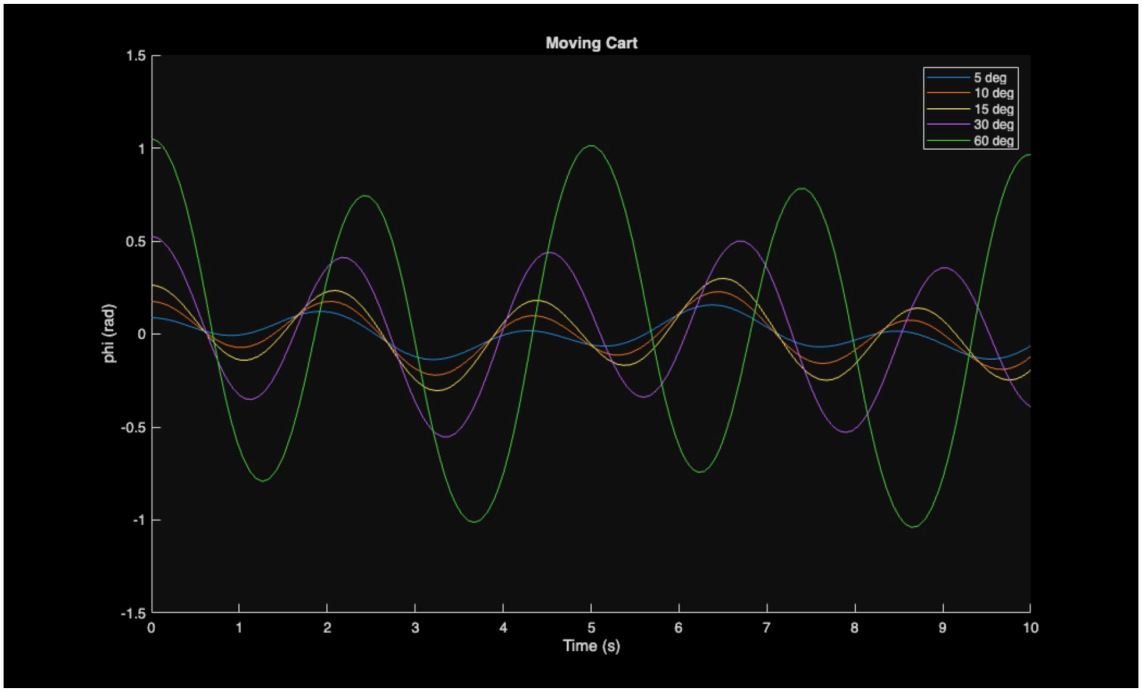
m = 1; % kg
l = 2; % m
k = 5; % N/m
M = 2; % kg
g = 9.81; % m/s2

fprintf("omega: %.2d rad/s\n", sqrt(g/l))
fprintf("period: %.2d s\n", 2 * pi / sqrt(g/l))

[t1,y1] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_1);
[t2,y2] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_2);
[t3,y3] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_3);
[t4,y4] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_4);
[t5,y5] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_5);
[t6,y6] = ode45(@(t,y) cart_moving_ode(t, y, l, g, m, M, k), ...
    tspan, cart_moving_initial_conditions_6);

figure(2)
hold on
plot(t2, y2(:,4))
plot(t3, y3(:,4))
plot(t4, y4(:,4))
plot(t5, y5(:,4))
plot(t6, y6(:,4))
hold off
title("Moving Cart")
legend('5 deg', '10 deg', '15 deg', '30 deg', '60 deg')
xlabel("Time (s)")
ylabel("phi (rad)")
```

Question assigned to the following page: 1.4



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